Introduction to Machine Learning

AI & Machine Learning Track

Mentor Session • Microsoft Campus Club (MCC-KFS)



Code of Conduct

General Rules

- **Respect First –** We are one family; respect and equality are essential.
- **Clear Communication –** Contact your responsible person for any issues.
- **Group Content –** No discussions about university or professors.
- **Neutrality –** No political or party-related talks.
- **Privacy –** No gossip or personal mentions inside or outside the club.

Code of Conduct

General Rules

- **Links –** No sharing personal/external links without approval.
- **Official Links –** Only share club links if instructed by admins.
- **Confidentiality –** Do not share internal data without authorization.
- **Hierarchy –** Escalate issues step by step: Mentor → Lead → Director → President.

Code of Conduct

Work Guidelines

- **Engagement –** Stay active in groups and social media; low engagement affects rewards.
- **Deadlines –** Submit all tasks on time; delays impact evaluation.
- **Team Spirit –** Work, learn, and share knowledge with others.
- **Events & Sessions –** Attendance on time is required.
- **Excuses –** Inform HR in advance if you cannot attend.

Why Machine Learning?

Motivation & Traditional Limits

- **The Rule Explosion Problem –** Complex problems (e.g. spam detection, facial recognition) require thousands of fragile if-else rules.
- **Fluctuating Environments –** Spammers change their words daily; hardcoded rules require constant, painful manual maintenance.
- **Human Intuition Bottleneck –** Humans easily recognize a handwritten digit '4', but describing exact pixel rules in code is nearly impossible.
- **The ML Solution –** Instead of hardcoding rules, we feed data to an algorithm that automatically discovers patterns and adapts over time.

Traditional Programming vs. Machine Learning

A Fundamental Paradigm Shift

Traditional Programming

Rules + Data → Answers

- **Logic Flow –** Developer writes explicit rules and logic in code.
- **Inputs –** Takes known Data and pre-programmed Rules.
- **Output –** Executes instructions to compute Answers.
- **Limitation –** Completely breaks down on complex non-linear tasks.
- **Best For –** Calculators, banking transactions, payroll logic.

Machine Learning

Data + Answers → Rules (Model)

- **Logic Flow –** Algorithm automatically discovers underlying rules.
- **Inputs –** Takes historical Data and known Answers (Labels).
- **Output –** Produces a trained Model (learned rules).
- **Advantage –** Easily handles unstructured data (images, text, audio).
- **Best For –** Spam filters, medical diagnosis, recommendation systems.

What is Machine Learning?

Core Concept & Intuition

- **Foundational Intuition –** Giving computers the ability to learn from historical data and past experience without being explicitly programmed.
- **Core Principle –** Algorithms analyze data to discover statistical patterns and build mathematical models that generalize to unseen inputs.
- **Formal Learning Criteria –** A computer program is said to learn when its measured performance on a specific task progressively improves through experience.
- **Concrete Spam Example –** Task: identify spam emails; Experience: analyzing thousands of labeled emails; Performance: percentage of accurately flagged emails.

When Should We Use Machine Learning?

Key Indicators & Decision Criteria

- **Long Hand-Tuned Rules –** When existing traditional solutions require huge lists of fine-tuned heuristics that are hard to maintain.
- **Complex Problems –** Problems for which traditional algorithmic approaches have no known solution (e.g., speech recognition, autonomous driving).
- **Fluctuating Environments –** Systems that must dynamically adapt to new data trends without manual software redeployment.
- **Extracting Insights (Data Mining) –** Discovering non-obvious patterns hidden in vast amounts of complex corporate or scientific data.
- **When NOT to Use ML –** Never use ML if a simple 5-line deterministic rule or database SQL query solves the problem reliably!

Core Terminology: The Data

Building Blocks of Every Dataset

- **Dataset –** The entire collection of structured or unstructured data points used for training and testing.
- **Instance / Sample –** A single observation or individual row in a dataset (e.g., one patient, one house, one email).
- **Feature (X) –** An individual measurable property or explanatory input column (e.g., area in sq ft, number of bedrooms, patient age).
- **Target / Label (y) –** The true outcome, ground truth, or dependent variable we want the model to learn to predict (e.g., house price, disease positive/negative).

Core Terminology: Model & Process

From Architecture to Prediction

- **Model –** The specific mathematical function or representation learned by the algorithm to map input features X to predictions.
- **Training / Fitting –** The process where the algorithm analyzes training data and repeatedly updates its internal parameters to minimize error.
- **Inference / Prediction –** Using the trained model on new, never-before-seen data to output an estimated target value (y_pred).
- **Ground Truth –** The verified real-world label used to benchmark how well the model performed.

Types of Machine Learning

The Two Primary Learning Paradigms

Supervised Learning

Data with Labels (X → y)

- **Core Idea –** Algorithm trains on inputs alongside their known answers (labels).
- **Analogy –** A student learning with an answer key and teacher corrections.
- **Goal –** Predict the correct target y for completely new input instances X.
- **Problem Types –** Regression (continuous) and Classification (discrete categories).
- **Examples –** House price estimation, spam classification, medical diagnosis.

Unsupervised Learning

Data without Labels (X only)

- **Core Idea –** Algorithm receives raw data without any target labels or guidance.
- **Analogy –** A child sorting mixed toys into natural clusters on their own.
- **Goal –** Discover hidden structural patterns, groups, or data anomalies.
- **Problem Types –** Clustering, Dimensionality Reduction, Anomaly Detection.
- **Examples –** Customer market segmentation, fraud anomaly detection.

Supervised Learning: Regression

Predicting Continuous Numerical Values

- **Core Question –** Answers quantitative questions: 'How much?' or 'How many?'
- **Target Variable –** The target (y) is a continuous number that can take any value within a range.
- **Real-World Example 1 –** Real Estate: Predicting the selling price of a house based on area, bedrooms, and location.
- **Real-World Example 2 –** Weather & Energy: Forecasting tomorrow's temperature or hourly electricity consumption.
- **Key Algorithms –** Linear Regression, Decision Tree Regressor, Random Forest Regressor.

Supervised Learning: Classification

Predicting Discrete Categories

- **Core Question –** Answers qualitative questions: 'Which category does this instance belong to?'
- **Target Variable –** The target (y) is a discrete class label from a predefined set of categories.
- **Binary Classification –** Exactly two classes: Spam vs. Not Spam, Fraud vs. Legitimate, Malignant vs. Benign.
- **Multiclass Classification –** Three or more classes: Handwritten digit recognition (0 through 9), sentiment analysis (Positive, Neutral, Negative).
- **Key Algorithms –** Logistic Regression, Support Vector Machines (SVM), Decision Trees.

Quick Interactive Check

Can You Identify the Problem Type?

- **Scenario A –** Predicting whether a bank loan applicant will default on payment (Yes or No)?
- **Scenario B –** Estimating the market salary of a software engineer based on years of experience and skill stack.
- **Scenario C –** Discovering natural purchasing personas among e-commerce customers without pre-existing labels.
- **Scenario D –** Forecasting tomorrow's peak electricity consumption in megawatts.



The Machine Learning Workflow

High-Level Project Lifecycle

Phase 1: Framing

Steps 1–3

- **Define Problem –** Formulate business objective.
- **Collect Data –** Gather historical records.
- **Understand Data –** Explore distributions & stats.

Phase 2: Prep

Steps 4–6

- **Data Cleaning –** Handle missing & noisy values.
- **Feature Prep –** Scale and encode variables.
- **Dataset Split –** Segregate Train, Val, Test.

Phase 3: Modeling

Steps 7–9

- **Select Model –** Pick baseline candidates.
- **Train Model –** Fit algorithms on train data.
- **Evaluate –** Assess errors on validation set.

Phase 4: Launch

Steps 10–12

- **Improve Model –** Fine-tune hyperparameters.
- **Final Testing –** One-time test set check.
- **Deploy & Monitor –** Serve predictions live.

Workflow: Framing & Data Understanding

Phase 1: Foundation of Every ML Project

1. Problem Definition

Step 01

- **Business Goal –** What real-world problem are we solving?
- **ML Framing –** Supervised or Unsupervised? Regression or Classification?
- **Success Metric –** Define business KPIs and evaluation metrics before coding.

2. Data Collection

Step 02

- **Data Sources –** Databases, APIs, sensors, web scraping, or system logs.
- **Data Quantity –** Ensure sufficient sample size for learning patterns.
- **Legal & Privacy –** Confirm compliance and user privacy standards.

3. Data Exploration

Step 03

- **Inspect Schema –** Identify data types, features X, and target y.
- **Summary Statistics –** Check mean, median, distributions, and skewness.
- **Detect Flaws –** Identify missing values, duplicates, and corrupt readings.

Workflow: Preparation & Splitting

Phase 2: Turning Raw Data into ML Ready Features

4. Data Cleaning

Step 04

- **Missing Values –** Drop rows or impute with median/mode.
- **Outliers –** Investigate and treat corrupt measurements.
- **Consistency –** Fix typographical errors in categories.

5. Feature Engineering

Step 05

- **Encoding –** Convert text categories to numbers (One-Hot / Ordinal).
- **Feature Scaling –** Standardize numeric columns to similar scales.
- **New Features –** Create domain ratios that help the model learn.

6. Dataset Splitting

Step 06

- **Segregation –** Split into Training, Validation, and Test sets.
- **Golden Rule –** Split BEFORE doing any preprocessing or fitting!
- **Leakage Defense –** Guarantees the model never peeks at future data.

Workflow: Training & Evaluation

Phase 3: Teaching and Validating the Model

7. Model Selection

Step 07

- **Algorithm Choice –** Select candidate baseline algorithms.
- **Simplicity First –** Start with simple models before complex ensembles.
- **Baseline Score –** Establish a benchmark to beat.

8. Model Training

Step 08

- **Fitting Weights –** Algorithm updates internal parameters on Train Set.
- **Minimizing Loss –** Optimizes objective loss function iteratively.
- **Check Convergence –** Ensure training stabilizes properly.

9. Validation Assessment

Step 09

- **Unseen Data –** Evaluate performance strictly on Validation Set.
- **Error Metrics –** Calculate RMSE / MAE or Accuracy / F1-Score.
- **Diagnose State –** Detect Overfitting vs. Underfitting.

Workflow: Improvement & Deployment

Phase 4: Optimization, Verification, and Production

10. Model Improvement

Step 10

- **Hyperparameter Tuning –** Systematically tune settings (Grid/Random Search).
- **Regularization –** Add constraints to prevent overfitting.
- **Ensemble Methods –** Combine predictions of multiple top models.

11. Final Test Check

Step 11

- **The Locked Vault –** Evaluate the single champion model on Test Set.
- **One-Time Evaluation –** Provides an unbiased estimate of real-world error.
- **Strict Warning –** NEVER tune hyperparameters on the Test Set!

12. Deployment & Monitoring

Step 12

- **Production Serving –** Deploy model via REST API or embedded service.
- **Drift Detection –** Monitor incoming live data for distribution shifts.
- **Continuous Retraining –** Retrain periodically as real-world trends evolve.

Dataset Splitting Strategy

The Essential Three-Way Division

Training Set

~70% of Total Data

- **Purpose –** Used exclusively to fit model parameters and learn patterns.
- **Analogy –** Study textbook and homework problems practiced during semester.
- **Access –** Seen repeatedly by the algorithm during the learning process.
- **Risk –** High score here does NOT guarantee real-world success (memorization).

Validation (Dev) Set

~15% of Total Data

- **Purpose –** Used to compare candidate models and tune hyperparameters.
- **Analogy –** Practice mock exams used to decide which study strategy works best.
- **Access –** Never used for parameter fitting; only for evaluation decisions.
- **Benefit –** Guides model selection without contaminating the test set.

Test Set

~15% of Total Data

- **Purpose –** Used once at the very end to estimate final real-world performance.
- **Analogy –** The final official certification exam taken only once.
- **Access –** Strictly locked away until the final model is fully selected.
- **Rule –** Never modify the model based on test set results!

Preventing Bias & Data Leakage

Why Two Splits Are Not Enough

- **The Overfitting Trap –** Evaluating on training data allows models to memorize samples rather than generalize to new data.
- **Information Peeking Bias –** Tuning hyperparameters directly on the test set compromises its role as an unbiased benchmark.
- **What is Data Leakage? –** When information from outside the training split unintentionally contaminates the training process.
- **Classic Leakage Trap –** Fitting feature scalers on the entire dataset before splitting. Always fit scalers ONLY on training data!
- **The Golden Principle –** Treat validation and test datasets as strictly unseen real-world instances at all times.

Challenges in Machine Learning

The Impact of Poor-Quality Data

- **Garbage In, Garbage Out –** Data riddled with noise, errors, and extreme outliers makes signal detection virtually impossible.
- **Insufficient Quantity –** Complex non-linear relationships require large volumes of representative observations to generalize.
- **Non-Representative Sampling –** If training data does not reflect production distributions, predictions will be inherently biased.
- **Measurement Errors –** Missing values, manual entry typos, and uncalibrated features severely degrade model performance.
- **Irrelevant Features –** Feeding uninformative columns introduces noise; thorough feature selection is essential.

Questions & Discussion

Let's Connect and Learn Together

- **Open Floor –** Feel free to ask any questions about today's concepts or problem formulations.
- **Community Channels –** Engage with your track peers and mentors on the official MCC WhatsApp and Discord.
- **Office Hours –** Reach out to your mentor for guidance on scenarios, intuition, or study materials.
- **Thank You! –** Great work today! See you in Week 2 as we dive deeper into Machine Learning!

